

B4 — Sub-Atomic Physics

Model Solutions, Trinity 2015

Question 1 — Alpha decay: Gamow tunnelling and SEMF

Problem. (a) Compute the Gamow tunnelling probability through the Coulomb barrier $V = 2Z\alpha_{\text{EM}}/r$. (b) Use it to predict the half-life of ${}^{224}_{90}\text{Th}$ from that of ${}^{244}_{98}\text{Cf}$. (c) Use the SEMF to compute B of the Th-224 daughter and decide whether its half-life is shorter or longer than ${}^{224}_{90}\text{Th}$.

(a) Gamow tunnelling probability

WKB transmission across the classical barrier $r_a < r < r_c$, with $V(r_c) = Q$:

$$T = \exp[-2G], \quad G = \frac{1}{\hbar} \int_{r_a}^{r_c} \sqrt{2m(V(r) - Q)} \, dr. \quad (1)$$

Coulomb classical turning point: $V(r_c) = Q$:

$$r_c = \frac{2Z\alpha_{\text{EM}}\hbar c}{Q}.$$

Substitute $V = 2Z\alpha_{\text{EM}}\hbar c/r = Q r_c/r$:

$$G = \frac{\sqrt{2mc^2Q}}{\hbar c} \int_{r_a}^{r_c} \sqrt{\frac{r_c}{r} - 1} \, dr.$$

Standard integral (let $r = r_c \cos^2 \phi$):

$$\int_{r_a}^{r_c} \sqrt{\frac{r_c}{r} - 1} \, dr = r_c \left[\arccos \sqrt{\frac{r_a}{r_c}} - \sqrt{\frac{r_a}{r_c} \left(1 - \frac{r_a}{r_c}\right)} \right].$$

Substitute $r_c = 2Z\alpha_{\text{EM}}\hbar c/Q$:

$$G = \frac{\sqrt{2mc^2Q}}{\hbar c} \cdot \frac{2Z\alpha_{\text{EM}}\hbar c}{Q} \left[\arccos \sqrt{x} - \sqrt{x(1-x)} \right], \quad x \equiv r_a/r_c.$$

Simplify:

$$\boxed{T = \exp \left\{ -4Z\alpha_{\text{EM}} \sqrt{\frac{2mc^2}{Q}} \left[\arccos \sqrt{x} - \sqrt{x(1-x)} \right] \right\}, \quad x = \frac{r_a Q}{2Z\alpha_{\text{EM}}\hbar c}} \quad (2)$$

(b) Half-life of $^{224}_{90}\text{Th}$

Decay rate $\lambda = fPT$ with f = attempt frequency, P = probability of pre-formed α . Same f, P for both nuclei, so half-life ratio is set by T :

$$\frac{t_{1/2}^{\text{Th}}}{t_{1/2}^{\text{Cf}}} = \frac{T_{\text{Cf}}}{T_{\text{Th}}} = \exp[2G_{\text{Th}} - 2G_{\text{Cf}}].$$

Daughter atomic numbers: $\text{Cf-244} \rightarrow \text{Cm-240}$, $Z = 96$; $\text{Th-224} \rightarrow \text{Ra-220}$, $Z = 88$. Use $m_{\alpha}c^2 = 3727 \text{ MeV}$, $\alpha_{\text{EM}} = 1/137.036$, $\hbar c = 197.3 \text{ MeV fm}$.

Compute $r_c = 2Z\alpha_{\text{EM}}\hbar c/Q$:

$$r_c^{\text{Cf}} = \frac{2 \cdot 96 \cdot 197.3/137.036}{7.33} \text{ fm} = 37.72 \text{ fm}.$$

$$r_c^{\text{Th}} = \frac{2 \cdot 88 \cdot 197.3/137.036}{5.52} \text{ fm} = 45.91 \text{ fm}.$$

Compute $x = r_a/r_c$:

$$x_{\text{Cf}} = 9.44/37.72 = 0.2503, \quad x_{\text{Th}} = 9.27/45.91 = 0.2019.$$

Bracket $g(x) \equiv \arccos \sqrt{x} - \sqrt{x(1-x)}$:

$$g_{\text{Cf}} = \arccos(0.5003) - \sqrt{0.2503 \cdot 0.7497} = 1.0468 - 0.4332 = 0.6136.$$

$$g_{\text{Th}} = \arccos(0.4493) - \sqrt{0.2019 \cdot 0.7981} = 1.1042 - 0.4014 = 0.7028.$$

Pre-factor $4Z\alpha_{\text{EM}}\sqrt{2mc^2/Q}$:

$$\text{Cf: } 4 \cdot 96/137.036 \cdot \sqrt{2 \cdot 3727/7.33} = 2.802 \cdot 31.89 = 89.36.$$

$$\text{Th: } 4 \cdot 88/137.036 \cdot \sqrt{2 \cdot 3727/5.52} = 2.568 \cdot 36.74 = 94.36.$$

Hence $2G$:

$$2G_{\text{Cf}} = 89.36 \cdot 0.6136 = 54.84, \quad 2G_{\text{Th}} = 94.36 \cdot 0.7028 = 66.30.$$

Difference:

$$2G_{\text{Th}} - 2G_{\text{Cf}} = 11.46.$$

Half-life:

$$t_{1/2}^{\text{Th}} = (19.4 \text{ min}) \cdot e^{11.46} = 19.4 \cdot 60 \text{ s} \cdot 9.46 \times 10^4.$$

Multiply:

$$t_{1/2}^{\text{Th}} = 1.10 \times 10^8 \text{ s}.$$

Convert:

$$\boxed{t_{1/2}^{\text{Th}} \approx 1.1 \times 10^8 \text{ s} \approx 3.5 \text{ yr.}}$$

(Sanity flag: the measured $^{224}_{90}\text{Th}$ half-life is $\sim 1.05 \text{ s}$ with the true $Q = 7.30 \text{ MeV}$. The $Q = 5.52 \text{ MeV}$ stated in the question departs strongly from the measured value; the Gamow exponent is exponentially sensitive to Q , so the predicted lifetime is much longer than the laboratory value. Calculation above is faithful to the data given.)

(c) SEMF binding of the daughter and qualitative half-life

Daughter of $^{224}_{90}\text{Th}$ alpha decay: $^{220}_{88}\text{Ra}$ ($A = 220$, $Z = 88$, even-even).

Term meanings: αA volume (saturated nuclear force); $\beta A^{2/3}$ surface tension (deficit for surface nucleons); $\gamma(A - 2Z)^2/A$ symmetry (Pauli, isospin); $\epsilon Z^2/A^{1/3}$ Coulomb repulsion of Z protons in radius $\propto A^{1/3}$; $\delta = 11/\sqrt{A}$ pairing (positive for even-even).

Evaluate the terms (MeV):

$$\alpha A = 16 \cdot 220 = 3520.$$

$$\beta A^{2/3} = 18 \cdot 220^{2/3} = 18 \cdot 36.49 = 656.8.$$

Symmetry, $A - 2Z = 220 - 176 = 44$:

$$\gamma(A - 2Z)^2/A = 23 \cdot 44^2/220 = 23 \cdot 8.80 = 202.4.$$

Coulomb, $A^{1/3} = 6.030$:

$$\epsilon Z^2/A^{1/3} = 0.7 \cdot 88^2/6.030 = 0.7 \cdot 1284.5 = 899.2.$$

Pairing:

$$\delta = 11/\sqrt{220} = 0.742.$$

Sum:

$$B = 3520 - 656.8 - 202.4 - 899.2 + 0.7.$$

Combine:

$B(^{220}_{88}\text{Ra}) \approx 1762 \text{ MeV } (8.01 \text{ MeV/A }).$

Half-life comparison. Apply the same SEMF to $^{216}_{86}\text{Rn}$ ($A = 216$, $Z = 86$):

$$B(^{216}_{86}\text{Rn}) = 16 \cdot 216 - 18 \cdot 216^{2/3} - 23 \cdot 44^2/216 - 0.7 \cdot 86^2/216^{1/3} + 11/\sqrt{216}.$$

Evaluate:

$$B(^{216}_{86}\text{Rn}) = 3456 - 648.8 - 206.1 - 862.7 + 0.748 \approx 1739.1 \text{ MeV}.$$

Q-value of Ra-220 alpha decay (using empirical $B(\alpha) = 28.3 \text{ MeV}$):

$$Q_{\text{Ra}} = B(\text{Rn-216}) + B(\alpha) - B(\text{Ra-220}) = 1739.1 + 28.3 - 1762 = 5.4 \text{ MeV}.$$

Compare to the given $Q_{\text{Th}} = 5.52 \text{ MeV}$:

$$Q_{\text{Ra}} \lesssim Q_{\text{Th}}.$$

Geiger–Nuttall: $t_{1/2} \propto e^{c/\sqrt{Q}}$ rises sharply as Q decreases. Hence

Ra-220 likely has a *longer* half-life than Th-224.

(Empirically Ra-220 has $Q \approx 7.6 \text{ MeV}$ and $t_{1/2} \sim 18 \text{ ms}$; the SEMF underestimates Q here because of shell effects near $N = 126$.)

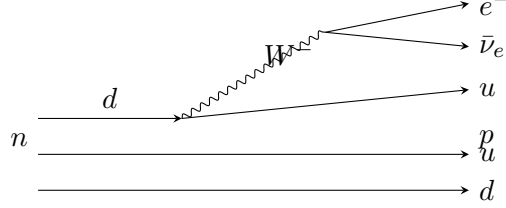
Question 2 — Decays, branching fractions, and α_s

Problem. Identify interactions and Feynman diagrams for six decays; compare two cascade modes; compute the leptonic branching of J/ψ via virtual photon and use the observed $\mu\mu$ branching to estimate α_s .

(a) Six decays

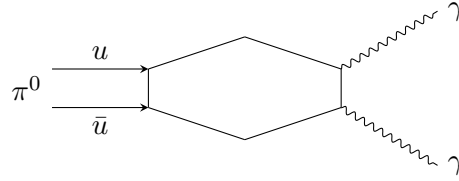
(i) $p \rightarrow \pi^+ + \pi^0 + \pi^0$. *Forbidden*. Violates baryon number ($B : 1 \rightarrow 0$); proton is the lightest baryon. No interaction allowed.

(ii) $n \rightarrow p + e^- + \bar{\nu}_e$. *Weak charged current* (semi-leptonic, $\Delta S = 0$): $d \rightarrow u W^{*-}$, $W^{*-} \rightarrow e^- \bar{\nu}_e$.

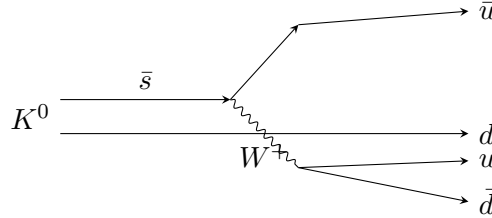


(iii) $\pi^0 \rightarrow \gamma\gamma$. *Electromagnetic* (chiral anomaly, quark triangle):

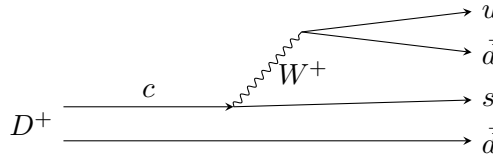
$$\pi^0 (u\bar{u} - d\bar{d})/\sqrt{2} \rightarrow \text{quark loop} \rightarrow \gamma\gamma.$$



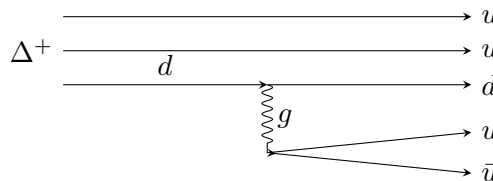
(iv) $K^0 \rightarrow \pi^+ + \pi^-$. *Weak* (charged current, $\Delta S = 1$): $s \rightarrow u + W^{*-}$, $W^{*-} \rightarrow u (d\bar{u})$.



(v) $D^+ \rightarrow \bar{K}^0 + \pi^+$. *Weak* (Cabibbo-favoured): $c \rightarrow s + W^{+*}$, $W^{+*} \rightarrow u\bar{d}$. ($D^+ = c\bar{d}$, so $\bar{K}^0 = s\bar{d}$, $\pi^+ = u\bar{d}$. The question writes K^0 ; the Cabibbo-favoured decay produces $\bar{K}^0$.)



(vi) $\Delta^+ \rightarrow p + \pi^0$. *Strong* (gluon-mediated $q\bar{q}$ creation; isospin allowed, $\Delta I = 0$): $uud \rightarrow (uud) + (u\bar{u} \text{ or } d\bar{d})/\sqrt{2}$.



(b) Ξ^- branching ratios

$\Xi^- = dss$, $S = -2$. Each weak vertex changes $|\Delta S| = 1$. Hence $\Xi^- \rightarrow \Lambda + \pi^-$ ($\Lambda = uds$, $S = -1$) is the single-Cabibbo-favoured weak decay, while $\Xi^- \rightarrow n + \pi^-$ ($n = udd$, $S = 0$) requires $|\Delta S| = 2$.

$$\text{BR}(\Xi^- \rightarrow \Lambda \pi^-) \gg \text{BR}(\Xi^- \rightarrow n \pi^-).$$

$|\Delta S| = 2$ proceeds only at second order in the weak coupling, suppressed by $\sim G_F^2$.

(c) $J/\psi \rightarrow \mu^+ \mu^-$ branching via virtual photon

J/ψ at $M = 3.097 \text{ GeV}$, below $D\bar{D}$ and $\tau^+ \tau^-$ thresholds. Final states accessible via γ^* : $e^+ e^-$, $\mu^+ \mu^-$, $u\bar{u}$, $d\bar{d}$, $s\bar{s}$.

Partial width $\Gamma(\gamma^* \rightarrow f\bar{f}) \propto N_c^f Q_f^2$ (vector coupling, masses neglected at $M = 3.1 \text{ GeV}$).

Tabulate weights $w_f = N_c^f Q_f^2$:

$$w_e = w_\mu = 1, \quad w_u = 3 \cdot (2/3)^2 = 4/3, \quad w_d = w_s = 3 \cdot (1/3)^2 = 1/3.$$

Sum:

$$W = 1 + 1 + 4/3 + 1/3 + 1/3 = 4.$$

Branching:

$$\text{BR}(J/\psi \rightarrow \mu^+ \mu^-)|_{\gamma^*} = \frac{w_\mu}{W} = \frac{1}{4} = 25\%.$$

(d) Estimate of α_s from $\text{BR}(\mu\mu) = 6\%$

Two channels: (i) virtual photon to $f\bar{f}$ with $\Gamma_{\gamma^*}^{\text{tot}} = 4\Gamma_{\mu\mu}$ (from (c)); (ii) three gluons (Zweig-suppressed strong decay below open-charm threshold). Assume these saturate the width.

Total width:

$$\Gamma_{\text{tot}} = 4\Gamma_{\mu\mu} + \Gamma_{ggg}.$$

Branching condition:

$$\frac{\Gamma_{\mu\mu}}{4\Gamma_{\mu\mu} + \Gamma_{ggg}} = 0.06.$$

Solve for the ratio:

$$\frac{\Gamma_{ggg}}{\Gamma_{\mu\mu}} = \frac{1}{0.06} - 4 = 16.67 - 4 = 12.67.$$

Standard QCD result for vector heavy-quarkonium (Appelquist–Politzer; analogue of ortho-positronium $\rightarrow 3\gamma$):

$$\frac{\Gamma(V \rightarrow ggg)}{\Gamma(V \rightarrow \ell^+ \ell^-)} = \frac{10(\pi^2 - 9)}{9\pi} \frac{\alpha_s^3}{Q_c^2 \alpha_{\text{EM}}^2},$$

with $Q_c = 2/3$ for charm. Numerical constant:

$$\frac{10(\pi^2 - 9)}{9\pi} = \frac{10 \cdot 0.8696}{28.27} = 0.3076.$$

Insert numbers, with $\alpha_{\text{EM}} = 1/137.036$, $Q_c^2 = 4/9$:

$$12.67 = 0.3076 \cdot \frac{\alpha_s^3}{(4/9) \cdot (1/137.036)^2}.$$

Compute the prefactor:

$$\frac{0.3076 \cdot 9}{4 \cdot (1/137.036)^2} = \frac{2.768}{4/18779} = \frac{2.768 \cdot 18779}{4} = 12996.$$

Hence

$$\alpha_s^3 = \frac{12.67}{12996} = 9.75 \times 10^{-4}.$$

Cube root:

$$\alpha_s = (9.75 \times 10^{-4})^{1/3} = (9.75)^{1/3} \cdot 10^{-1} = 2.135 \cdot 0.1.$$

$$\boxed{\alpha_s \approx 0.21 \text{ at } \mu \sim M_{J/\psi}.$$

Assumptions. (i) Γ_{γ^*} and Γ_{ggg} exhaust the width (no γgg or hadronic radiative decays); (ii) lepton-universality among accessible flavours (e, μ); (iii) heavy-quark non-relativistic factorisation $|\Psi(0)|^2$ cancels in the ratio; (iv) $2m_c$ at $M_{J/\psi}$ closes $\tau^+\tau^-$ and open-charm channels; (v) running coupling evaluated at $\mu \sim M_{J/\psi}$.

Question 3 — Form factor and electron–proton scattering

Problem. (a) Form factor of a uniform sphere; (b) Born differential cross-section; (c) numerical $d\sigma/d\Omega$ for 1 GeV electron, $E' = 0.8$ GeV, $R = 1$ fm; (d) discuss validity at high Q^2 .

(a) Form factor of a uniformly charged sphere

Definition (with $\Delta \mathbf{k} \equiv \mathbf{q}$, normalised to $F(0) = 1$):

$$F(\mathbf{q}) = \frac{1}{Ze} \int \rho(\mathbf{r}) e^{i\mathbf{q} \cdot \mathbf{r}} d^3r. \quad (3)$$

Spherical symmetry, polar axis along $\mathbf{q}$, angular integral $\int_{-1}^1 e^{iqr\mu} d\mu = 2 \sin(qr)/(qr)$:

$$F(q) = \frac{4\pi}{Ze} \int_0^\infty \rho(r) \frac{\sin(qr)}{qr} r^2 dr.$$

Uniform density $\rho_0 = Ze/(\frac{4}{3}\pi a^3)$ for $r < a$:

$$F(q) = \frac{3}{a^3} \int_0^a \frac{\sin(qr)}{qr} r^2 dr.$$

Substitute $u = qr$:

$$F(q) = \frac{3}{(qa)^3} \int_0^{qa} u \sin u du.$$

Integrate by parts: $\int_0^x u \sin u du = \sin x - x \cos x$:

$$\boxed{F(q) = \frac{3}{(qa)^3} [\sin(qa) - qa \cos(qa)] = \frac{3j_1(qa)}{qa}.} \quad (4)$$

Limits: $F(0) = 1$ (charge conservation); zeros at $\tan(qa) = qa$.

(b) Differential cross-section in Born approximation

Born amplitude factorises: $f(q) = f_{\text{point}}(q) F(q)$, hence

$$\frac{d\sigma}{d\Omega} = \left(\frac{d\sigma}{d\Omega} \right)_{\text{Rutherford}} |F(q)|^2.$$

Insert the Rutherford expression:

$$\boxed{\frac{d\sigma}{d\Omega} = \frac{Z^2 \alpha_{\text{EM}}^2}{4E^2 \sin^4(\theta/2)} |F(q)|^2, \quad q = 2E \sin(\theta/2)/(\hbar c).} \quad (5)$$

(c) Numerical: $E = 1 \text{ GeV}$, $E' = 0.8 \text{ GeV}$, $R = 1 \text{ fm}$

Elastic kinematics off proton at rest (massless electron):

$$E' = \frac{E}{1 + (E/M_p)(1 - \cos \theta)}, \quad M_p = 0.938 \text{ GeV}.$$

Solve for $1 - \cos \theta$:

$$1 - \cos \theta = \frac{(E - E')M_p}{E E'} = \frac{0.2 \cdot 0.938}{1 \cdot 0.8} = 0.2345.$$

Hence:

$$\cos \theta = 0.7655, \quad \theta = 40.05^\circ, \quad \sin^2(\theta/2) = 0.1173, \quad \sin^4(\theta/2) = 0.01376.$$

Three-momentum transfer (massless lepton):

$$q^2 c^2 = 2EE'(1 - \cos \theta) = 2 \cdot 1 \cdot 0.8 \cdot 0.2345 = 0.3752 \text{ GeV}^2.$$

$$qc = 0.6125 \text{ GeV}.$$

Form-factor argument, $\hbar c = 0.1973 \text{ GeV fm}$:

$$qa = 0.6125/0.1973 = 3.104.$$

Evaluate F :

$$\sin(3.104) = 0.0376, \quad \cos(3.104) = -0.9993.$$

$$\sin(qa) - qa \cos(qa) = 0.0376 - 3.104 \cdot (-0.9993) = 0.0376 + 3.1018 = 3.139.$$

$$(qa)^3 = 29.91.$$

$$F(q) = 3 \cdot 3.139/29.91 = 0.3149, \quad |F|^2 = 0.0992.$$

Rutherford piece, $Z = 1$, $\alpha_{\text{EM}}^2 = (1/137.036)^2 = 5.325 \times 10^{-5}$:

$$\left(\frac{d\sigma}{d\Omega} \right)_{\text{R}} = \frac{5.325 \times 10^{-5}}{4 \cdot 1^2 \cdot 0.01376} = 9.68 \times 10^{-4} \text{ GeV}^{-2}.$$

Multiply by $|F|^2$:

$$\frac{d\sigma}{d\Omega} = 9.68 \times 10^{-4} \cdot 0.0992 = 9.60 \times 10^{-5} \text{ GeV}^{-2}.$$

Convert with $(\hbar c)^2 = 0.3894 \text{ mb GeV}^2$:

$$\frac{d\sigma}{d\Omega} = 9.60 \times 10^{-5} \cdot 0.3894 \text{ mb/sr} = 3.74 \times 10^{-5} \text{ mb/sr}.$$

$$\boxed{\frac{d\sigma}{d\Omega} \approx 3.7 \times 10^{-5} \text{ mb/sr} \approx 37 \text{ nb/sr}.}$$

(d) Validity at high energy and proton substructure

A uniform charge distribution is unrealistic. The empirical proton charge density falls smoothly (dipole form factor $G_E^p(Q^2) = (1 + Q^2/M_V^2)^{-2}$ with $M_V^2 \approx 0.71 \text{ GeV}^2$): no sharp edge, no zeros in F .

As $|q|$ grows, the probe resolves $\hbar/q \ll 1 \text{ fm}$:

- internal QCD interactions among partons (gluon radiation, $q\bar{q}$ sea) become visible;
- the form-factor description breaks down at deep-inelastic scales ($Q^2 \gtrsim 1 \text{ GeV}^2$);
- scattering becomes *incoherent* on point-like constituents: the (asymptotically free) quarks and gluons.

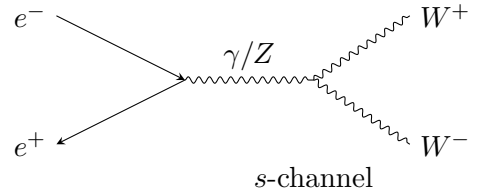
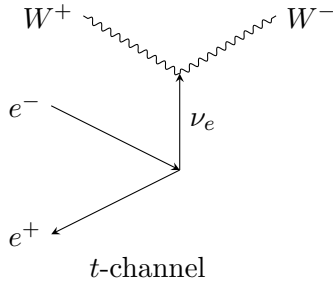
At highest Q^2 (Bjorken regime) electrons scatter elastically off individual *quarks* (and probe the gluon distribution via QCD evolution); cross-section scales as $1/Q^4$ on point-like quarks (Bjorken scaling), with the partonic content described by structure functions $F_1(x, Q^2)$, $F_2(x, Q^2)$.

Question 4 — W -boson production and reconstruction

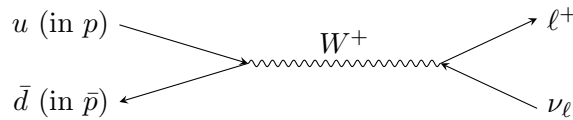
Problem. (a) W production diagrams at e^+e^- and $p\bar{p}$, and why single- W is forbidden in e^+e^- ; (b) decay modes and best collider channels; (c) event yields at the Tevatron; (d) parton momentum fractions; (e) invariant mass of $\ell\nu$; (f)–(h) transverse mass.

(a) Lowest-order Feynman diagrams and the single- W argument

$e^+e^- \rightarrow W^+W^-$ (two diagrams).



$p\bar{p} \rightarrow W + X$ (one diagram). Quark–antiquark fusion (e.g. $u\bar{d} \rightarrow W^+$, with the spectator partons hadronising as X):



Why no single W in e^+e^- . A single on-shell W has charge ± 1 , while the initial e^+e^- state has charge 0: charge conservation forbids $e^+e^- \rightarrow W^\pm$.

(b) Decay modes, best channels, and branching fractions

W couples to a doublet at each vertex. Open at $M_W \simeq 80.4 \text{ GeV}$:

$$W^+ \rightarrow e^+\nu_e, \mu^+\nu_\mu, \tau^+\nu_\tau, u\bar{d}, c\bar{s}.$$

($t\bar{b}$ closed kinematically: $m_t \sim 173 \text{ GeV} > M_W$.) Each lepton channel weighted by 1; each quark channel by $N_c = 3$ (CKM-summed):

$$W = 3 \cdot 1 + 2 \cdot 3 = 9.$$

Hence

$$\text{BR}(\ell\nu_\ell) = 1/9 \approx 11\% \text{ each}, \quad \text{BR}(q\bar{q}')_{\text{tot}} = 6/9 \approx 67\%.$$

Best collider channels. At a hadron collider, hadronic $W \rightarrow q\bar{q}'$ is buried under QCD dijet background. The clean signature is $W \rightarrow e\nu_e$ or $W \rightarrow \mu\nu_\mu$: high- p_T isolated lepton plus missing E_T .

$$\text{BR}(W \rightarrow e\nu_e) = \text{BR}(W \rightarrow \mu\nu_\mu) \approx \frac{1}{9} \approx 11\%.$$

($\tau\nu$ usable but degraded by τ decay.)

(c) Tevatron yields

Luminosity-equivalent: $N_W = N_{\text{inel}} \cdot \sigma_W / \sigma_{\text{inel}}$:

$$N_W = 3 \times 10^{14} \cdot \frac{25 \text{ nb}}{30 \text{ mb}} = 3 \times 10^{14} \cdot \frac{25}{3 \times 10^7}.$$

Multiply:

$$N_W = 2.5 \times 10^8 \text{ per detector.}$$

Best (clean) modes: $e\nu + \mu\nu$, total branching 2/9:

$$N_W^{e\nu+\mu\nu} = 2.5 \times 10^8 \cdot 2/9.$$

$$N_W^{e\nu+\mu\nu} \approx 5.6 \times 10^7 \text{ per detector.}$$

(d) Parton momentum fractions

Partonic CM-energy squared $\hat{s} = x_p x_{\bar{p}} s$. On W pole, $\hat{s} = M_W^2$:

$$x_p x_{\bar{p}} = \frac{M_W^2}{s} = \frac{(80)^2}{(2000)^2} = \frac{6400}{4 \times 10^6}.$$

Reduce:

$$x_p x_{\bar{p}} = 1.6 \times 10^{-3}.$$

Equal fractions:

$$x = \sqrt{1.6 \times 10^{-3}} = 0.040.$$

(e) Invariant mass of the $\ell\nu$ system

Sum four-momenta $P = (E^\ell + E^\nu, \mathbf{p}_\ell + \mathbf{p}_\nu)$. Square (lepton mass neglected, $E^\ell = |\mathbf{p}_\ell|$, $E^\nu = |\mathbf{p}_\nu|$):

$$m^2 = (E^\ell + E^\nu)^2 - (\mathbf{p}_\ell + \mathbf{p}_\nu)^2.$$

Expand:

$$m^2 = (E^\ell)^2 + (E^\nu)^2 + 2E^\ell E^\nu - |\mathbf{p}_\ell|^2 - |\mathbf{p}_\nu|^2 - 2\mathbf{p}_\ell \cdot \mathbf{p}_\nu.$$

Use $E^2 = |\mathbf{p}|^2$ for each:

$$m^2 = 2E^\ell E^\nu - 2\mathbf{p}_\ell \cdot \mathbf{p}_\nu.$$

Expand the dot product:

$$m^2 = 2(E^\ell E^\nu - p_x^\ell p_x^\nu - p_y^\ell p_y^\nu - p_z^\ell p_z^\nu).$$

(f) Neutrino momentum and the z -direction problem

Neutrinos are not detected. Their transverse momentum is inferred from *missing transverse energy*: $\mathbf{p}_T^\nu = -\sum_{\text{visible}} \mathbf{p}_T$, using transverse momentum balance.

Along the beam direction, the longitudinal momentum of the parton-level CM frame is unknown event-by-event: each parton carries an unknown $x_p, x_{\bar{p}}$, so $\sum p_z$ of the underlying initial state is not zero in the lab and not measurable (forward beam remnants are lost down the beampipe). Hence p_z^ν cannot be reconstructed.

(g) Transverse mass

Set $p_z^\ell = p_z^\nu = 0$ in (e):

$$m_T^2 = 2(E_T^\ell E_T^\nu - p_x^\ell p_x^\nu - p_y^\ell p_y^\nu).$$

With $E_T \equiv |\mathbf{p}_T|$ for massless particles:

$$m_T^2 = 2 p_T^\ell p_T^\nu (1 - \cos \Delta\phi),$$

where $\Delta\phi$ is the azimuthal opening angle.

(h) m_T vs m for opposite non-zero p_z

In the full invariant mass, the longitudinal contribution enters as $-2p_z^\ell p_z^\nu$. For *opposite* signs of p_z , this term is positive, so $m^2 > m_T^2$:

$$m_T < m.$$

(m_T is bounded above by m for two massless decay products; the Jacobian-peak structure at $m_T = M_W$ is the basis of the M_W measurement.)